

CHAPTER-18 (MANUAL-17)

Sewerage Operation Department

Other Useful information

About 3850 M.L.D. potable water is supplied to the city of Greater Mumbai every day and about 2727 MLD of sewage is generated. The sewage so generated needs to be collected and disposed off in such a manner so as to protect the ecological balance. It is also mandatory for corporation to abide by provisions of Prevention of Water Pollution Act, Environmental Pollution Act etc.

Sewerage Operations Department of M.C.G.M. operates and maintains the Sewerage Network, Sewage Pumping Stations, Sewage Treatment Plants(STP) and Waste Water Treatment Facilities (WWTF) brought in place by Sewerage Project and Mumbai Sewage Disposal Project departments in seven Sewerage Zones of Mumbai City. The sewage is conveyed through existing underground Sewer Network, collected at various Satellite Pumping Stations, treated at eight different Sewage Treatment Plants and disposed into nearby Sea/Creeks.

The department is also responsible for repairs and rehabilitation of existing sewer lines. The department has installed the sewage recycling plants of smaller capacity in the premises of sewage pumping stations for the reuse of sewage water for non-potable purposes. The department also provides transport services to various sections of Water Supply and Sewerage Disposal department.

The Department falls under the Budget 'G' of the M.C.G.M.

Departmental Activities:

Sewerage operation department is mainly divided into following sections according to the nature of work.

- i) Main sewer and pipe sewer section responsible for conveyance of Sewage
- ii) Pumping section responsible for collection, treatment and disposal of sewage
- iii) Transport section responsible for providing transport services to various sections of W.S.S.D. dept.
- iv) Civil section responsible for repairs to sewer lines & civil structures of S.O. dept.
- v) Planning & Construction section responsible for rehabilitation of Sewer lines, rising mains of pumping stations.

The main sewer & pipe sewer section undertakes yearly programme for systematic de-silting and cleaning of the sewer lines of the entire sewer network of Mumbai City for smooth flow of Sewage thereby minimizing the complaints of sewage overflow on the roads.

To keep the sewer lines in smooth working condition, the sewer lines are periodically inspected, desilted and cleaned at pre-decided intervals under a systematic cleaning programme. To prevent direct contact with sewage, the cleaning work is mechanized for which various sewer cleaning machines/equipments like Rodding Machines, Power Bucket Sewer Cleaning Machines, Suction Machines, High Capacity Suction Machines, Jetting Machines, High Capacity Suction and Jetting Machine with Filtration and Recycling System, Grab Bucket Machines etc. as detailed below are used with all safety norms.

- 1) Jetting machine : 05 Nos.
- 2) Suction Machine 6.5m³capacity : 11 Nos.
- 3) Suction Machine 1.5m³capacity : 20 Nos.
- 4) Power Bucket Machine : 22 Pairs
- 5) Choke Machine(Power Rodding) : 26 Nos.
- 6) High Capacity Suction And Jetting : 10 Nos.
Machine with Filtration & Recycling
System
- 7) Grab type Manhole Desilting : 08 Nos.
Machine

The department has three centralized control rooms one each at City, Eastern and Western suburb equipped with Power Rodding Machines and other equipments. These control rooms work round the clock, 24x7 to attend and redress all complaints received pertaining to the sewerage system.

Any complaint about the non-functioning of the sewerage system is registered either with the concerned ward or with any of the three main sewer control rooms as stated above. The complaints are also recorded through the MCGM web site or on the central complaint number 1916. Most of the complaints received are resolved within 24 hours.

The sewage collected through the sewerage network spread all over Mumbai is conveyed to the terminal Pumping Station and treatment facilities through a network of sewers and satellite pumping stations for treatment and disposal.

There are seven sewage zones in Mumbai. Each of seven sewage zones has a terminal sewage pumping station and sewage treatment plant where the sewage is treated and further disposed into sea/creek. The sewage treatment plant at Colaba, Love Grove, Worli, Bandra provides primary treatment and the sewage discharged into Arabian Sea through Marine Outfalls. Similarly the sewage at Malad is discharged into Malad Creek after primary

treatment. The sewage treatment plants at Versova, Bhandup and Ghatkopar, provides secondary treatment through aerated lagoons and treated sewage is discharged into Malad & Thane creek.

The statistics of Sewerage Operation Department

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| 1) | Total Length of Sewer Lines | : | 2000 km of various sizes digitized in GIS |
| 2) | Number of Manholes | : | 72950 |
| 3) | Total Number of Street Connections | : | 348672(surveyed on May 2017) |
| 4) | Total Sewage Pumping Stations | : | 48 Nos. |
| 5) | Installed Capacity of Sewage Treatment Plants | : | 5041 M.L.D. |
| 6) | Permissible Disposal | : | 2727.4 MLD |
| 7) | Actual Disposal (Average) | : | 1851 MLD |
| 8) | Total Strength of Employees | | |
| | i) Sewerage Operations Division | : | 6557 Nos. |
| | ii) Transport Division | : | 788 Nos. |